Parity Assessment Report

Generated: **2026-04-09T23:54:48+00:00** (UTC)

Executive parity snapshot

Status: Tier checks and collected lane evidence currently support parity on this corpus.

Key outcomes

- Tier outcomes: 26/26 passed, 0 failed, 0 skipped.
- Coverage map: 14 instrumented, 0 pending, 0 excluded.
- Scalar evidence spans 15 path/condition lanes; Pearson metrics are present in 15 lanes.

Caveats

- Intrinsic ratios use pooled intrinsic MAE in the denominator; near-zero baselines can inflate ratios for ligand-autoregressive, tied-positions-and-multi-state [reference_weighted_sum__jax_product].
- Side-chain-conditioned integration gate reports 2 warning verdict(s), driven by excluded corpus lanes.

How parity is assessed

1. `parity_fast`: deterministic fixture and helper-branch parity checks (no upstream checkout required).
2. `parity_heavy`: reference-backed numerical parity checks against pinned LigandMPNN.
3. `parity_audit`: converted checkpoint-family load/audit checks across model families.

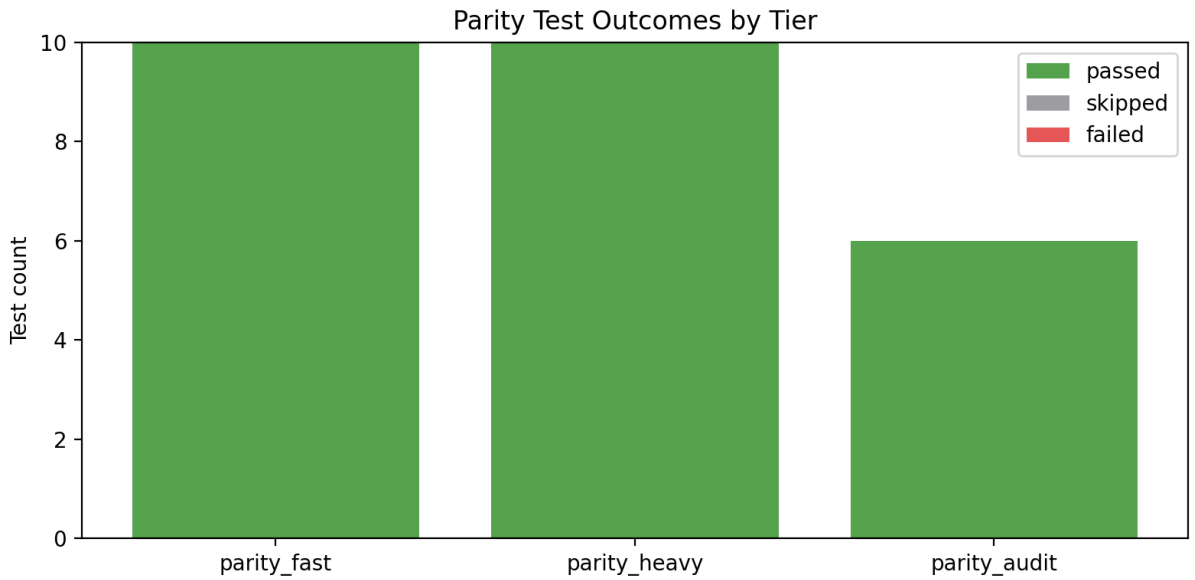
Terminology / interpretation legend

Term	Interpretation
Mean r / Min r	

Term	Interpretation
	Pearson correlation summaries per path/condition lane; values nearer 1 indicate tighter parity.
Pearson pass rate	Fraction of per-case Pearson checks passing policy thresholds for that lane.
n/a(categorical | pending | excluded)	Metric intentionally absent because the path is categorical, not instrumented, or excluded.
Cross/intrinsic ratio	Cross-model MAE divided by pooled intrinsic MAE; near-zero denominators can inflate ratios.
Coverage heatmap values	Binary semantics only: 1 means at least one evidence row is present; 0 means absent.

Tier test outcomes (from JUnit)

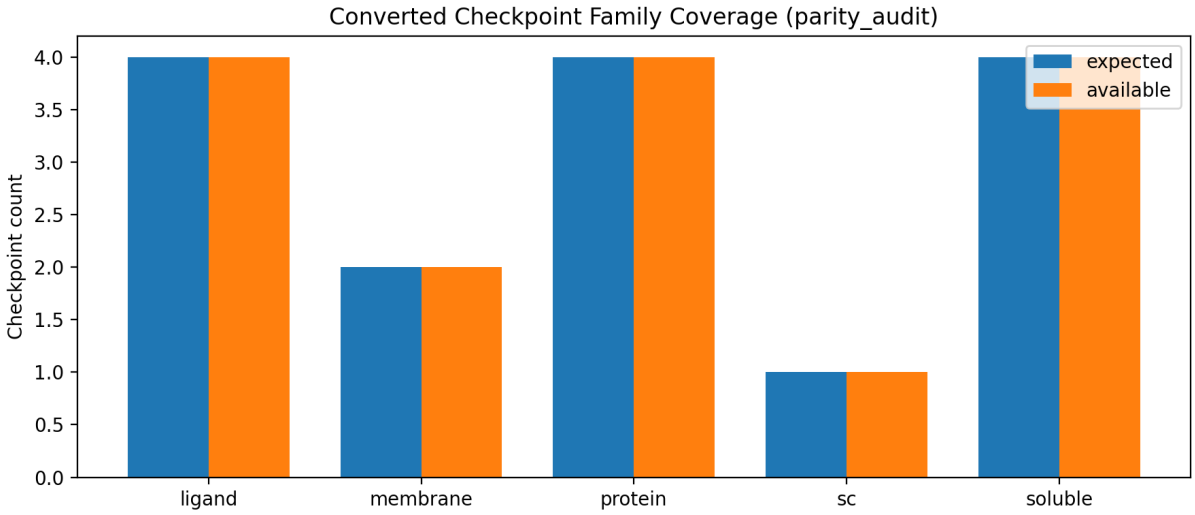
Tier	Tests	Passed	Skipped	Failed	Notes
`parity_fast`	10	10	0	0	
`parity_heavy`	10	10	0	0	
`parity_audit`	6	6	0	0	



Stacked JUnit pass/skip/fail totals per tier; this is the first release gate for triage.

Converted family checkpoint coverage

Family	Expected converted	Available locally
`ligand`	4	4
`membrane`	2	2
`protein`	4	4
`sc`	1	1
`soluble`	4	4



Expected vs locally present converted checkpoints for parity_audit; missing assets block audit evidence collection.

Expanded evidence summary

Evaluated protein systems and cases

Case metadata below is extracted directly from collected evidence rows.

Case ID	Kind	Length(s)	Seed(s)	Backbone ID(s)	Checkpoint ID(s)
`1ubq`	`real_backbone`	76	388	1ubq	ligandmpnn_sc_v_3 ligandmpnn_v_32_(proteinmpnn_v_48_

Case ID	Kind	Length(s)	Seed(s)	Backbone ID(s)	Checkpoint ID(s)
`5awl`	`real_backbone`	10	388	5awl	ligandmpnn_sc_v_3 ligandmpnn_v_32_(proteinmpnn_v_48_
`ligand`	`audit_family`	n/a	n/a	ligand	converted_checkpo
`membrane`	`audit_family`	n/a	n/a	membrane	converted_checkpo
`parity-fast-contract`	`synthetic`	76	489	synthetic	proteinmpnn_v_48_
`protein`	`audit_family`	n/a	n/a	protein	converted_checkpo
`sc`	`audit_family`	n/a	n/a	sc	converted_checkpo
`soluble`	`audit_family`	n/a	n/a	soluble	converted_checkpo

Tied/multistate apples-to-apples lanes

Configured tied/multistate lanes are listed beside evidence tables so unlike strategies never merge.

Condition	Comparison API	Reference combiner
`reference_weighted_sum__jax_product`	`sampling`	`weighted_su`
`reference_arithmetic_mean__jax_arithmetic_mean`	`scoring`	`arithmetic_`

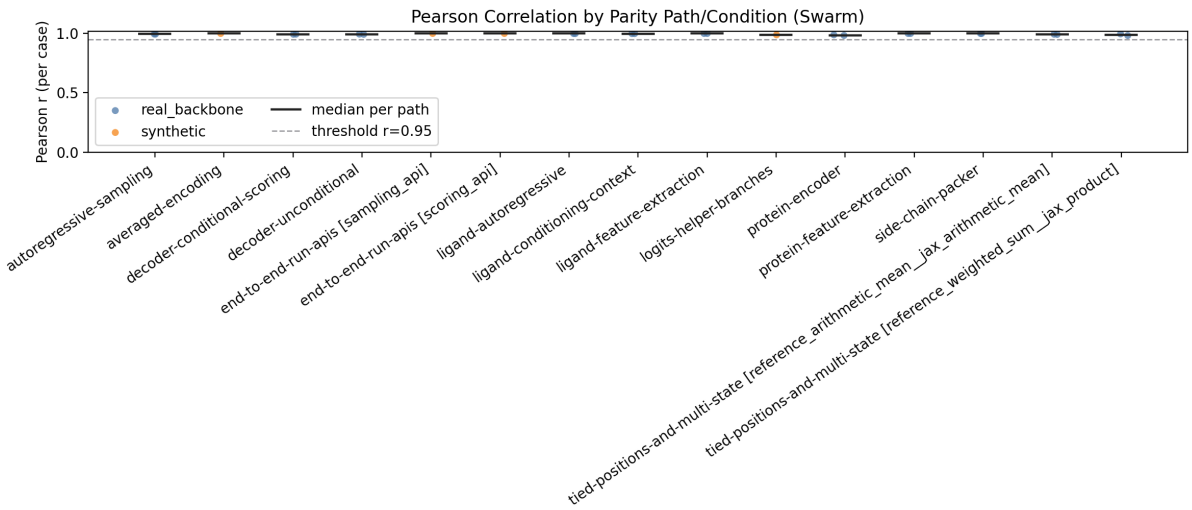
Evidence metrics: `docs/parity/reports/evidence/evidence_metrics.csv`
Point samples: `docs/parity/reports/evidence/evidence_points.csv`

Condition labels isolate strategy lanes (for example, tied/multistate apples-to-apples comparisons).

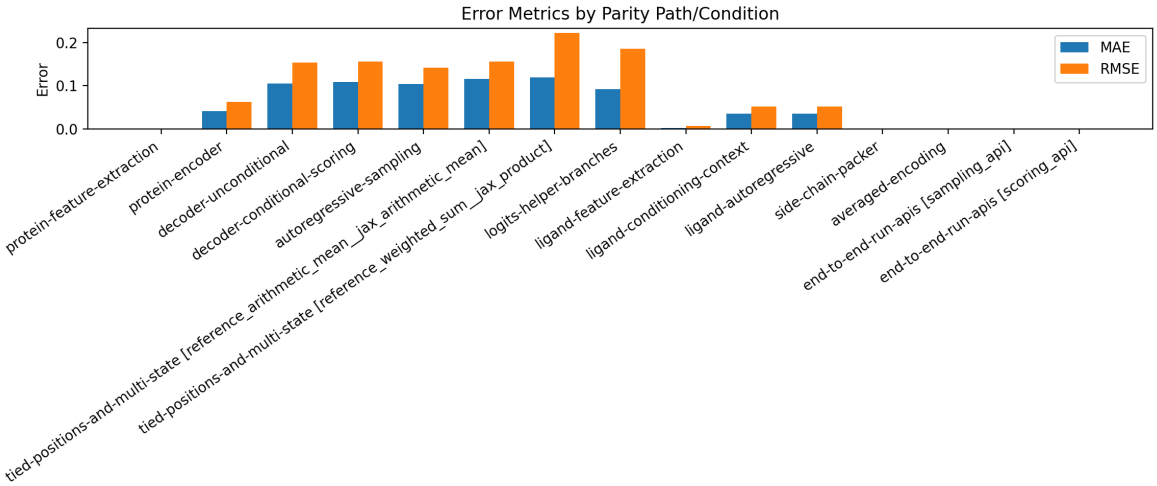
Path	Condition	Cases
	`default`	2

Path	Condition	Cases
`protein-feature-extraction`		
`protein-encoder`	`default`	2
`decoder-unconditional`	`default`	2
`decoder-conditional-scoring`	`default`	2
`autoregressive-sampling`	`default`	2
`autoregressive-sampling`	`no_ligand`	2
`tied-positions-and-multi-state`	`reference_arithmetic_mean__jax_arithmetic_mean`	2
`tied-positions-and-multi-state`	`reference_weighted_sum__jax_product`	2
`logits-helper-branches`	`default`	1
`ligand-feature-extraction`	`default`	2
`ligand-conditioning-context`	`default`	2
`ligand-autoregressive`	`default`	2
`ligand-autoregressive`	`ligand_context`	2
`ligand-autoregressive`	`side_chain_conditioned`	2

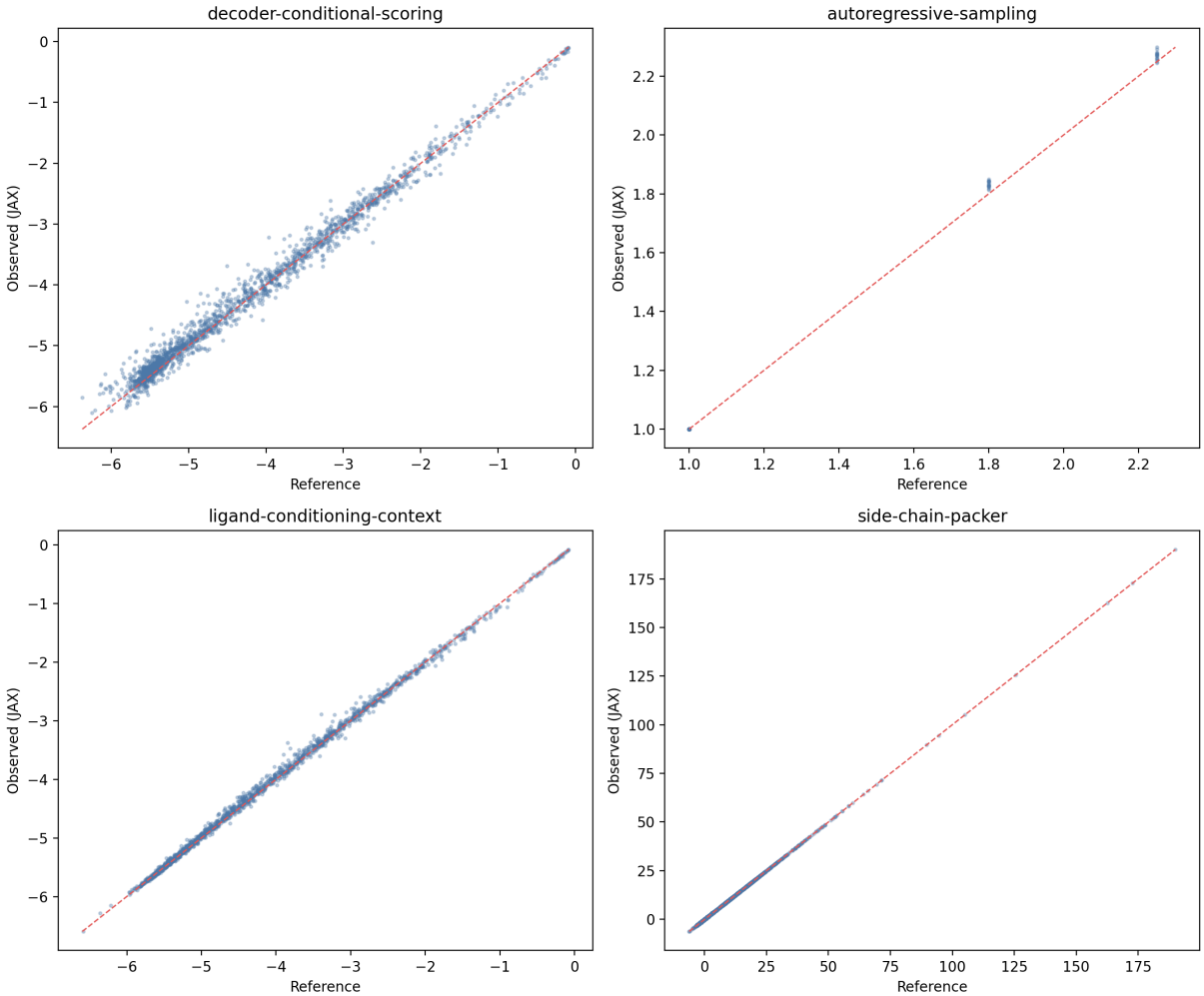
Path	Condition	Cases
`side-chain-packer`	`default`	2
`averaged-encoding`	`default`	1
`end-to-end-run-apis`	`sampling_api`	1
`end-to-end-run-apis`	`scoring_api`	1
`checkpoint-family-load`	`ligand`	1
`checkpoint-family-load`	`membrane`	1
`checkpoint-family-load`	`protein`	1
`checkpoint-family-load`	`sc`	1
`checkpoint-family-load`	`soluble`	1



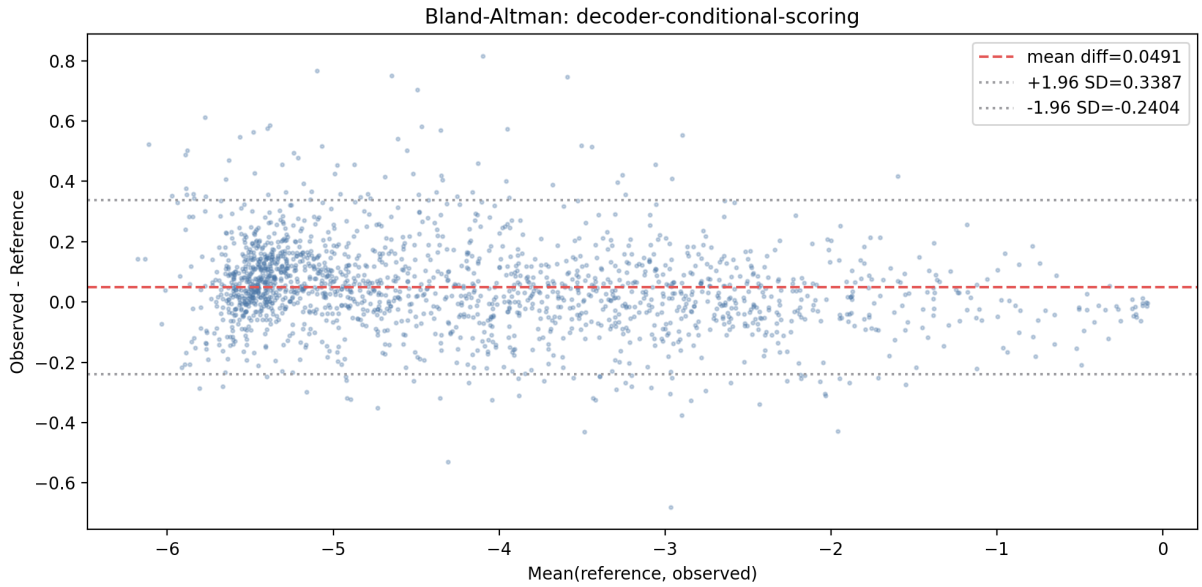
Pearson swarm across lane-level case rows (n=26). Each point is one path/condition/case/seed value, black ticks mark medians, and the dashed line marks the r=0.95 policy floor.



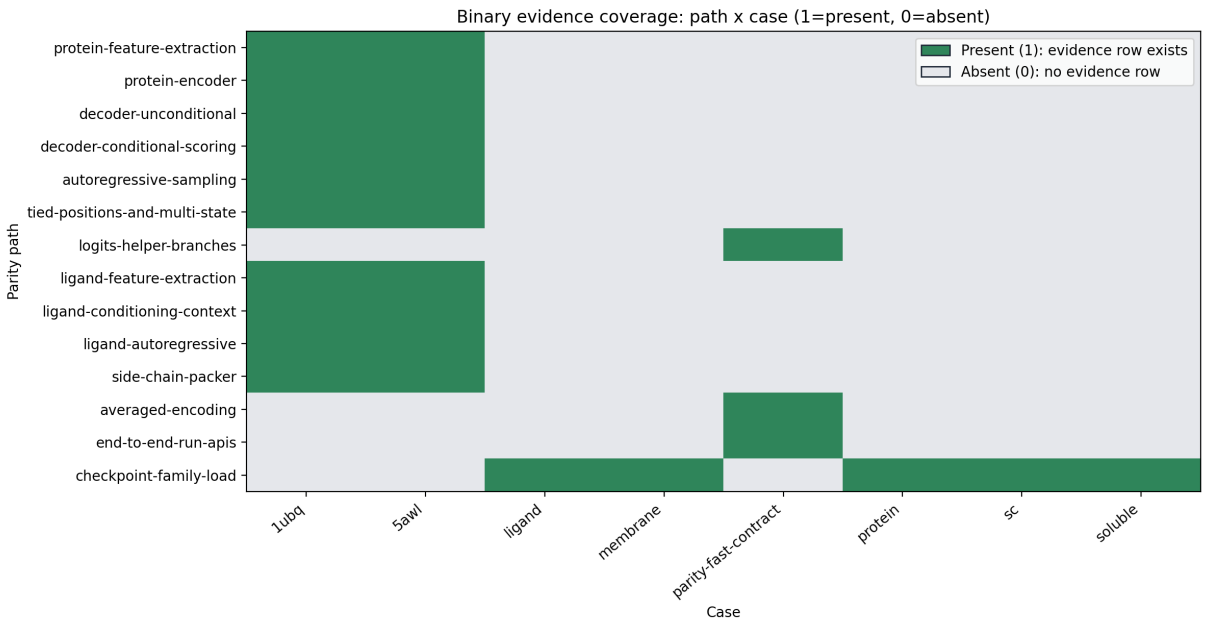
Mean MAE and RMSE across 15 scalar-metric lanes. Rows shown as n/a(...) are pending, excluded, or categorical by design.



Reference vs JAX scatter samples for core paths; tighter diagonal alignment indicates stronger parity. Dense clouds may be downsampled for readability.



Bland-Altman view for decoder-conditional-scoring. The center line is mean drift and dotted lines show ±1.96 SD.

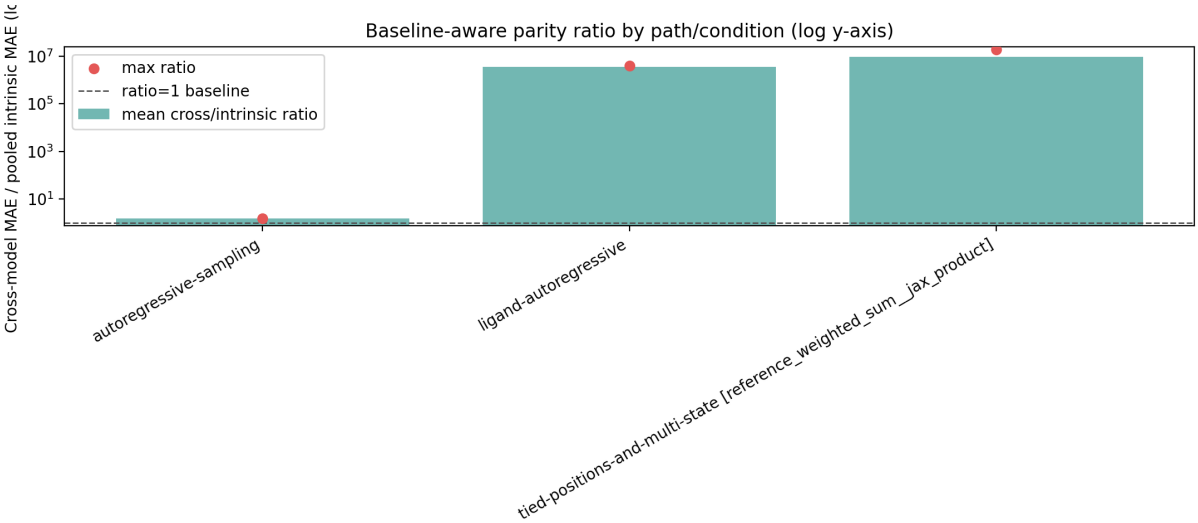


Binary coverage matrix (14 paths × 8 cases): green=present (>=1 evidence row), gray=absent. Color intensity does not encode magnitude.

Baseline-aware intrinsic parity

Path	Condition	Cases	Mean cross intrinsic ratio
`autoregressive-sampling`	`default`	2	1.5317

Path	Condition	Cases	Mean cross intrinsic ratio
`ligand-autoregressive`	`default`	2	3502778.82
`tied-positions-and-multi-state`	`reference_weighted_sum__jax_product`	2	9363044.01



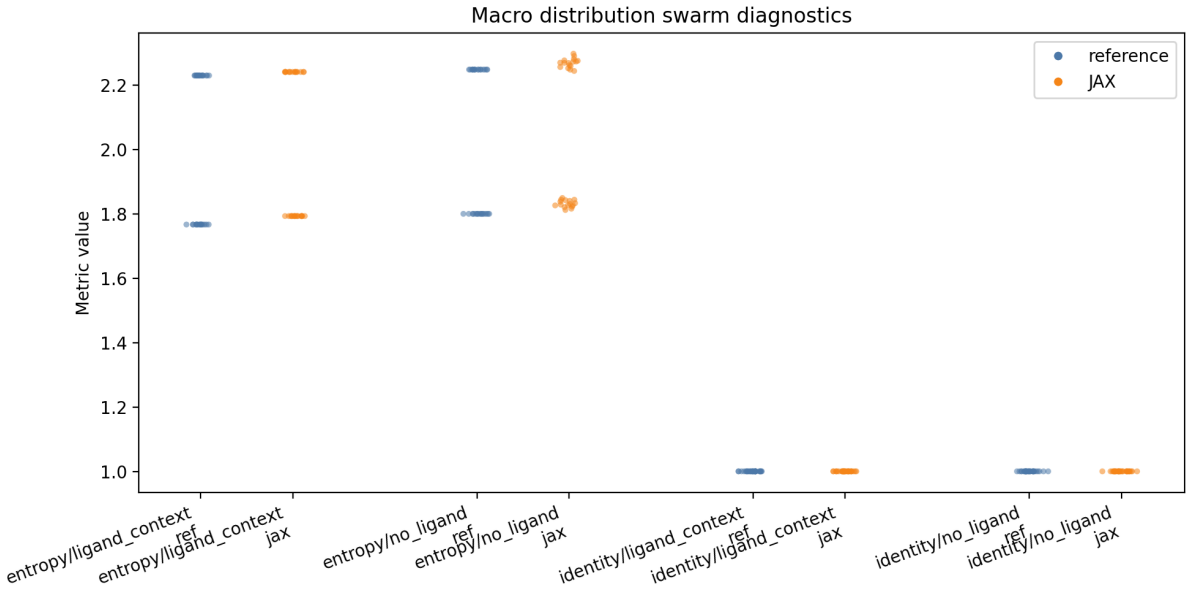
Log-scale cross/intrinsic ratio chart across 3 lane(s). The denominator is pooled intrinsic MAE, so near-zero baselines can inflate ratios.

Macro distribution parity

Path	Condition	Cases	Mean identity Wasserstein	Mean entropy Wasserste
`autoregressive-sampling`	`no_ligand`	2	0.0000	0.0261
`ligand-autoregressive`	`ligand_context`	2	0.0000	0.0187
`ligand-autoregressive`	`side_chain_conditioned`	2	n/a(Excluded by parity case corpus configuration.)	n/a(Exclude by parity ca corpus configuratio

Side-chain-conditioned integration gate

Path	Condition	Cases	Pass	Warn	Fail	Outcom
`ligand-autoregressive`	`side_chain_conditioned`	2	0	2	0	`warn`



Macro swarm compares reference vs JAX identity/entropy points by condition. Overlap implies better distributional parity; excluded lanes remain n/a(...).

Coverage and exclusions

Path	Tier	Status	Reason
`protein-feature-extraction`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`protein-encoder`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`decoder-unconditional`	`parity_heavy`	`instrumented`	Path has collected evidence rows.

Path	Tier	Status	Reason
`decoder-conditional-scoring`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`autoregressive-sampling`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`tied-positions-and-multi-state`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`logits-helper-branches`	`parity_fast`	`instrumented`	Path has collected evidence rows.
`ligand-feature-extraction`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`ligand-conditioning-context`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`ligand-autoregressive`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`side-chain-packer`	`parity_heavy`	`instrumented`	Path has collected evidence rows.
`averaged-encoding`	`parity_fast`	`instrumented`	Path has collected evidence rows.
`end-to-end-run-apis`	`parity_fast`	`instrumented`	Path has collected evidence rows.
`checkpoint-family-load`	`parity_audit`	`instrumented`	Audit path is categorical; correlation and

Path	Tier	Status	Reason
			error columns are intentionally n/a.

Appendix: path inventory and definitions

Parity path definitions

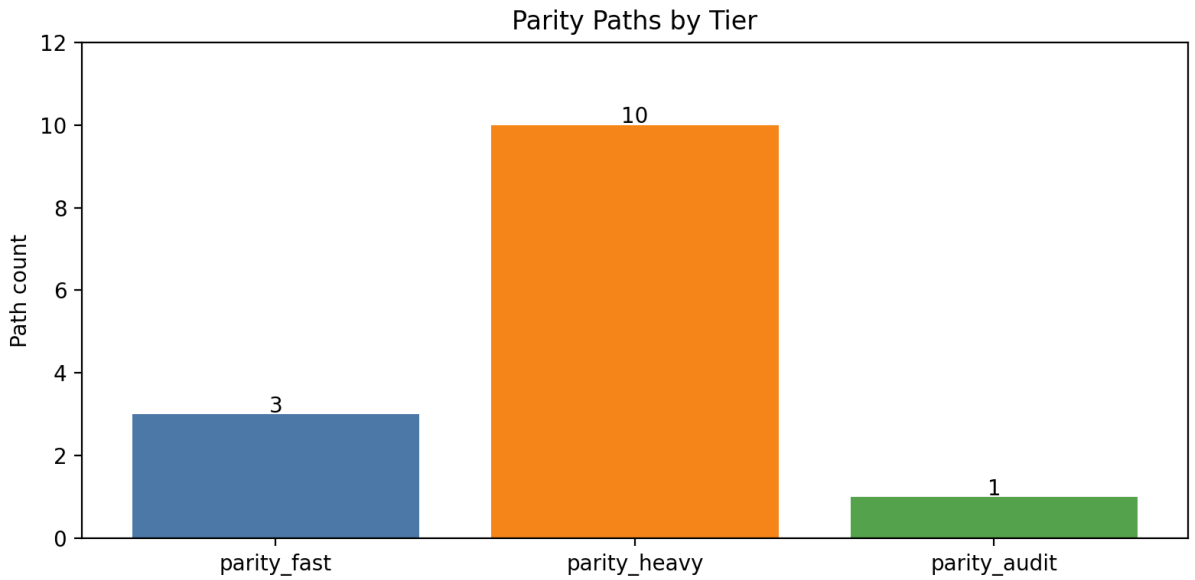
Path	Tier	Method	Metrics	Code paths
`protein-feature-extraction`	`parity_heavy`	intermediate tensor parity	shape equality, allclose, max abs diff	`src/prxteinmpnn features.py`
`protein-encoder`	`parity_heavy`	layer-by-layer activation parity	correlation, max abs diff, per-layer trace	`src/prxteinmpnn encoder.py`
`decoder-unconditional`	`parity_heavy`	end-logit parity	pearson correlation, per-position KL, per-position CE	`src/prxteinmpnn decoder.py, src prxteinmpnn/mod mpnn.py`
`decoder-conditional-scoring`	`parity_heavy`	logits and score parity	pearson correlation, score delta, nll drift	`src/prxteinmpnn decoder.py, src prxteinmpnn/sco score.py, src/ prxteinmpnn/run scoring.py`
`autoregressive-sampling`	`parity_heavy`	stepwise deterministic sample parity	stepwise logit drift, token agreement, sequence recovery	`src/prxteinmpnn sampling/sample prxteinmpnn/run sampling.py, sr prxteinmpnn/mod mpnn.py`

Path	Tier	Method	Metrics	Code paths
`tied-positions-and-multi-state`	`parity_heavy`	group-combined branch parity with explicit apples-to-apples comparison lanes	group correlation, strategy invariants	`src/prxteinmpnn` `autoregression.` `prxteinmpnn/mod` `multi_state_sam` `src/prxteinmpnn` `sampling/sample`
`logits-helper-branches`	`parity_fast`	helper vs direct branch parity	allclose, shape equality	`src/prxteinmpnn` `sampling/` `unconditional_l` `src/prxteinmpnn` `sampling/` `conditional_log`
`ligand-feature-extraction`	`parity_heavy`	ligand node/ edge parity	shape equality, masked behavior	`src/prxteinmpnn` `ligand_features`
`ligand-conditioning-context`	`parity_heavy`	context encoder parity	hidden-state correlation, final logit correlation	`src/prxteinmpnn` `mpnn.py`
`ligand-autoregressive`	`parity_heavy`	deterministic ligand sample parity	stepwise agreement, final agreement	`src/prxteinmpnn` `mpnn.py`
`side-chain-packer`	`parity_heavy`	encode/ decode torsion parity	mean allclose, concentration allclose, mix logits allclose, output correlation	`src/prxteinmpnn` `packer.py`

Path	Tier	Method	Metrics	Code paths
`averaged-encoding`	`parity_fast`	deterministic averaged encoding parity	shape equality, average drift	`src/prxteinmpnn averaging.py`
`end-to-end-run-apis`	`parity_fast`	fixture-driven API parity	output shape, score contract, pseudo-perplexity contract	`src/prxteinmpnn scoring.py, src prxteinmpnn/run sampling.py`
`checkpoint-family-load`	`parity_audit`	load and smoke forward pass	load success, smoke pass, shape compatibility	`src/prxteinmpnn weights.py, src prxteinmpnn/mod`

Tier path inventory

Tier	Path count
`parity_fast`	3
`parity_heavy`	10
`parity_audit`	1



Inventory-only path counts by tier. Larger bars indicate broader instrumentation, not stronger parity.

Checkpoint checksum snapshot: `reports/converted_checkpoint_checksums.txt`